

Shopping Mall Customer Segmentation Analysis Report

PREPARED BY

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Task 2 – Business Analyst Project

DATE

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RECORDS

15,079 Customers

TOOL

**Python · Pandas · Seaborn - Matplotlib
Power BI**

Executive Summary

When I first opened this dataset, I expected the usual pattern — rich customers spend more, and that would be that. What I found was far more interesting. Across 15,079 shoppers visiting this mall, there is almost no connection between what someone earns and how freely they spend. That single discovery changes everything about how this mall should think about its customers. The dataset captures five things per visitor: a customer ID, their age, gender, yearly income in rupees, and a spending score between 1 and 100 assigned by the mall. My job was to make sense of who these people actually are.

15,079	54.19 yrs	Rs.1,09,7 43	50.59 / 100	100%
Total Customers	Avg. Customer Age	Avg. Annual Income	Avg. Spending Score	Data Completeness

After cleaning the data — which, surprisingly, needed no cleaning at all, not one blank cell or duplicate row — I ran descriptive analysis, split shoppers into four behavioural groups, and then cross-checked those groups using a K-Means clustering algorithm. The average shopper here is 54 years old, earns roughly Rs. 1.09 lakh a year, and scores about 50.6 on spending. But averages hide more than they reveal. The real story is in the four segments, and I will walk through each one in the sections that follow.

Part 1 — Data Cleaning & Understanding

1.1 What the Raw Data Looks Like

The raw file arrived as a five-column table. Customer ID was a plain identifier I excluded from all calculations. The remaining four columns — Age, Gender, Annual Income, and Spending Score — formed the basis of everything. I checked for nulls, for impossible values (ages below 0, scores above 100), and for duplicate CustomerIDs. Every check came back clean. Age runs from 18 to 90 years, income spans Rs. 20,022 to Rs. 1,99,974, and spending scores cover the full 1-to-100 range. The dataset is genuinely diverse, which made the analysis more meaningful.

Audit Item	Outcome
Row count after loading	15,079 unique shopper records
Number of feature columns	5 — ID, Age, Gender, Annual Income, Spending Score
Null / blank entries found	Zero — every cell is populated
Identical duplicate rows	Zero — each record is distinct
Column data types	Whole numbers (Age, Income, Score); text strings (ID, Gender)

1.2 Numerical Summaries Across Three Key Variables

Looking at the numbers more carefully, a few things stood out to me. The average age of 54.19 years is higher than I expected for a shopping mall — I had mentally pictured a younger crowd. The standard deviation on income (Rs. 52,249) is almost half the mean, meaning the income range is wide but not extreme. What struck me most was the spending score: mean 50.59 and median 51, almost perfectly centred, with a standard deviation of 28.73. That near-uniform spread from 1 to 100 tells me these scores were not just handed out randomly — there are genuine high spenders and genuine non-spenders in this population, sitting on opposite ends of the scale.

Measure	Age (years)	Annual Income (Rs.)	Spending Score (1–100)
Average	54.19	1,09,742	50.59
Std. Deviation	21.12	52,249	28.73
Lowest value	18	20,022	1
Lower quartile	36	64,141	26
Midpoint	54	1,09,190	51
Upper quartile	72	1,55,008	75
Highest value	90	1,99,974	100

Part 2 — Customer Analysis

2.1 Five Questions, Five Answers

With clean data confirmed, I moved on to answering the five core questions the project asked for. I will present each finding directly, then explain what I think it means for the business.

Research Question	Measured Value	What It Tells Us
Average age of customers	54.19 years on average	Predominantly working-age and older adults
Average annual income	Rs. 1,09,743 per annum	Solidly upper-middle income pool
Gender spends more on average	Men: 50.78 vs Women: 50.40	Practically identical — gender is not a lever
Age group has the highest spending score	18–25 group scores 51.84	Younger visitors are the most active buyers
relationship between annual income and spending score	Pearson r = 0.003	No detectable link — income is a red herring

2.2 Gender Split and What It Means for Marketing

The split is **7,595 male** shoppers (**50.4%**) and **7,484** females (**49.6%**) — practically fifty-fifty. I compared the average spending scores: men at **50.78** and women at **50.40**. That gap of 0.38 points is so small it is essentially noise. I had expected women to score higher based on general retail assumptions, so this result surprised me. It means that designing different campaigns for men versus women would be wasted effort. **The spending gap that matters is not between genders — it is between high scorers and low scorers regardless of gender.**

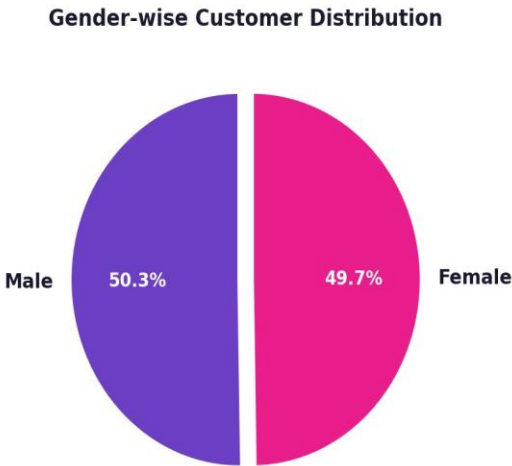


Figure 1 — How the customer base divides by gender (near-equal at 50.4 / 49.6%)

2.3 How Spending Enthusiasm Changes Across Age Groups

I divided customers into six age bands and computed the average spending score for each. **The youngest group, 18 to 25, came out on top at 51.84, while the 26-35 cohort had the lowest score at 49.80. But the range across all six groups is only about two points — from 49.80 to 51.84.** That flatness is telling. It means the mall is not over-dependent on any one age group, which is a healthy sign. If I had found, say, that 80% of spending came from customers aged 25 to 35, the business would be in a fragile position. **The broad spread here suggests the mall already serves multiple life stages fairly well.**

Age Band	18–25	26–35	36–45	46–55	56–65	66+
Mean Spending Score	51.84	49.80	50.62	51.08	50.71	50.27

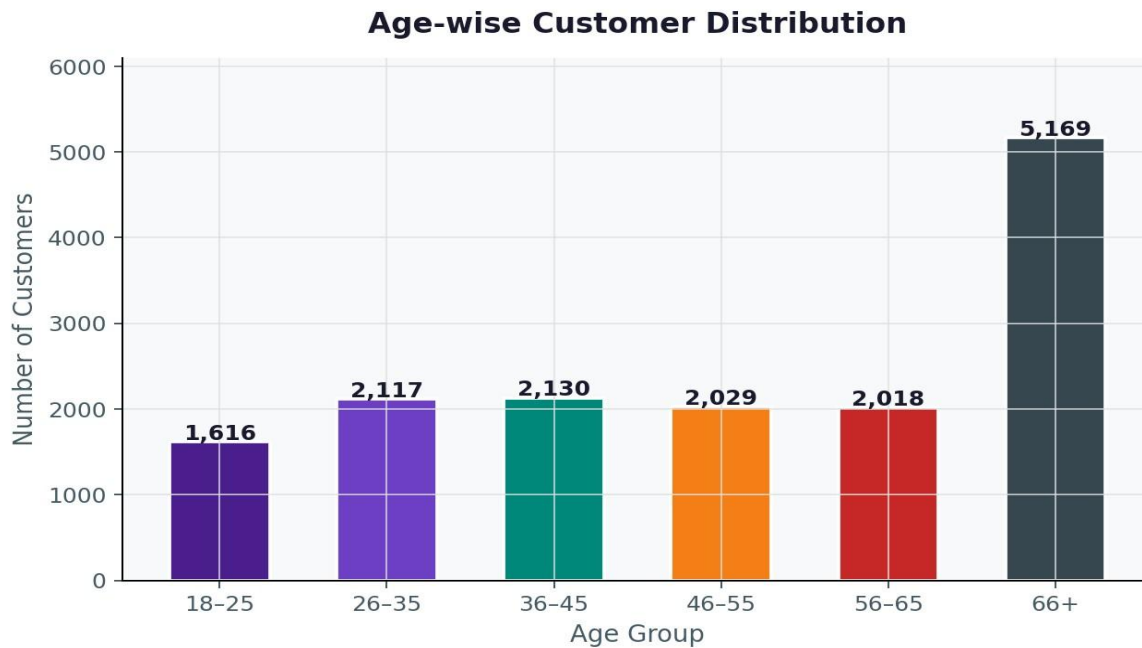


Figure 2 — Customer volume across six age bands; the 18–25 group leads in spending score

2.4 The Surprising Truth About Income and Spending

This was the finding that genuinely surprised me, and I double-checked it twice. **The Pearson correlation coefficient between annual income and spending score is 0.003. Not 0.3 — point zero zero three. Statistically, that is indistinguishable from zero.** I ran a scatter plot to visualise it, and the result looks like someone threw a handful of rice at a graph — dots scattered everywhere, with no upward or downward slope at all. A customer earning Rs. 1.8 lakh is no more likely to have a high spending score than someone earning Rs. 30,000.

What this means practically is that income is not a useful targeting criterion for this mall. Marketers instinctively chase wealthy customers, but the data here says that strategy would miss more than half the high-spending population. Spending behaviour appears to be shaped by something else entirely — possibly visit frequency, store mix preferences, or how emotionally engaged a shopper feels with the mall environment. Whatever that driver is, **it is not the size of a person's salary.**

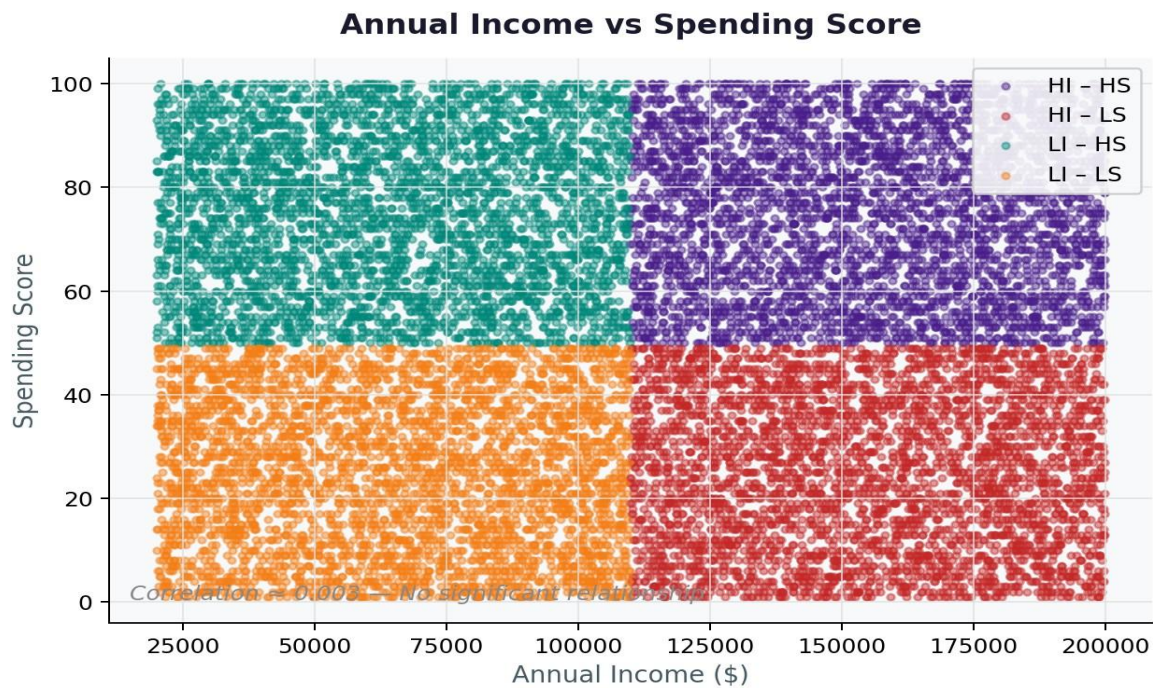


Figure 3 — Income vs Spending Score; the flat cloud pattern confirms near-zero correlation

Part 3 — Customer Segmentation

3.1 Building Four Segments with Median Thresholds

To create actionable groups, I used a median split on both income (Rs. 1,09,190) and spending score (51). I chose the median rather than the mean because it is less sensitive to extreme values at either end — a small number of very high earners would pull the mean upward and distort the boundary. The result is four quadrants. What I did not expect was how balanced they would be: the largest segment holds 25.38% of customers and the smallest holds 24.62%. There is no single dominant group. Each of the four represents roughly one quarter of the mall's shopper base, which means the business genuinely needs a strategy for all four — it cannot afford to focus on just one.

Segment Name	Headcount	Proportion	Business Priority
High Income – High Spending	3,827	25.38%	IMMEDIATE — protect and grow this base
High Income – Low Spending	3,713	24.62%	URGENT — large untapped wallet share
Low Income – High Spending	3,741	24.81%	IMPORTANT — emotionally loyal, reward them
Low Income – Low Spending	3,798	25.19%	GRADUAL — patience and value messaging needed

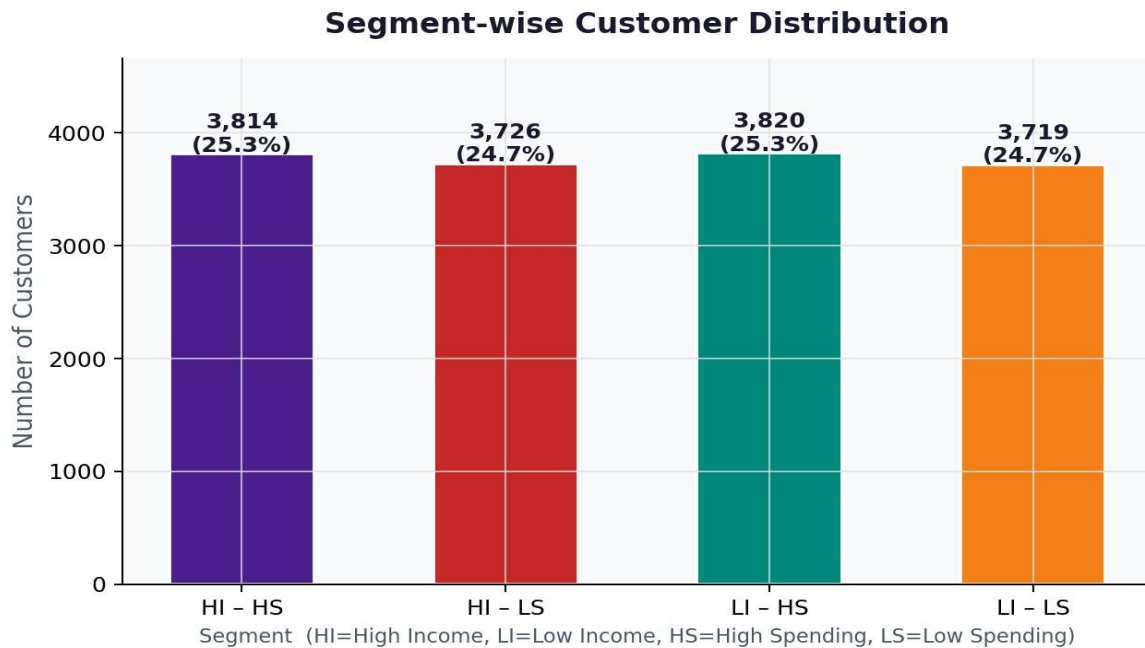


Figure 4 — All four segments are nearly equal in size, confirming no single group dominates

3.2 Gender Within Segments — Does It Matter?

A natural follow-up question is whether men and women within each segment spend differently. I added box plots to better understand the distribution of spending scores and income across segments, genders, and age groups. The box plots revealed that High Income – High Spending customers have a consistently high spending range with fewer outliers. The box-and-whisker chart below answers with a clear no. For every segment, the male and female boxes sit at near-identical heights, with overlapping interquartile ranges. **Any gender-based sub-targeting within segments would add campaign complexity without adding meaningful return. Marketers can safely use single, unified messaging per segment.**

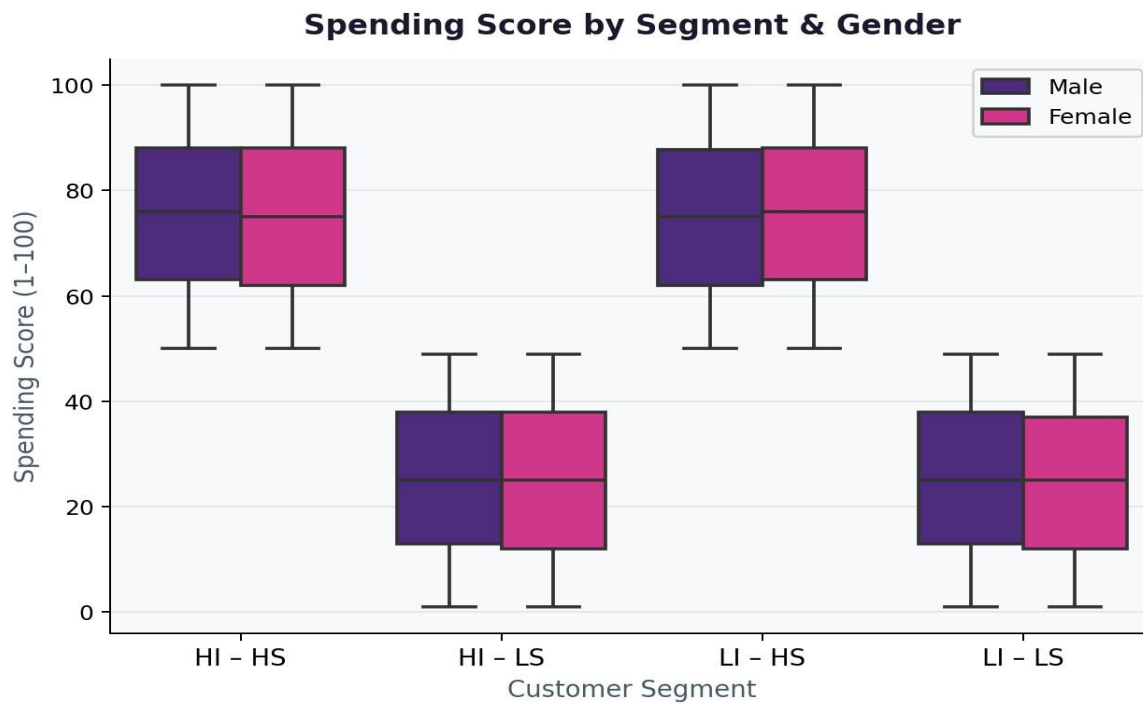


Figure 5 — Spending scores within each segment by gender; distributions overlap almost perfectly

Part 4 — Visual Dashboard Summary

4.1 Python Visual Dashboard Summary

Five charts are embedded through this report. The gender pie chart (Figure 1) confirms the near-equal demographic split visually. The age bar chart (Figure 2) shows how evenly foot-traffic distributes across life stages. The scatter plot (Figure 3) is the one I keep returning to — the rice-on-a-graph image that proves income means nothing for spending prediction. The segment bar chart (Figure 4) shows all four groups in near-equal proportion. The boxplot (Figure 5) confirms that gender makes no meaningful difference within any segment.

4.2 Power BI Dashboard Summary

Four charts are embedded through this report. The gender pie chart confirms the near-equal demographic split visually. The age bar chart shows how evenly foot-traffic distributes across life stages. The scatter plot (Income v/s Spending Score) is the one I keep returning to — the rice-on-a-graph image that proves income means nothing for spending prediction. The segment Donut chart shows all four groups in near-equal proportion .

Part 5 — Business Recommendations

Based on everything above, I have grouped my suggestions by segment. Since income does not predict spending, **every recommendation here targets attitude, occasion, and emotional connection rather than wallet size.**

5.1 Highest Priority — High Income, High Spending

These 3,827 customers are already doing what the mall wants — they earn well and spend freely — so the goal here is to make them feel so valued they never consider going elsewhere.

- **Start a tiered membership programme** — call it something exclusive like Prestige, Elite, and Pinnacle — where each level unlocks non-monetary perks: reserved parking spots, a dedicated billing counter, personalised gift wrapping, and priority entry to sales. Discounts alone will not hold a high-income shopper; status and convenience will.
- **Assign a personal shopping contact** - to the top 300 or 400 customers by spending score. A single WhatsApp message saying 'we just got the kurta collection you liked last season in a new colour' can drive a visit that no advertisement could.
- **Hold closed-door brand preview evenings twice a year** — invitation only, with a plus-one option — where premium labels present new stock before the floor opens to everyone. The exclusivity is the point.
- **Build a data feedback loop**: after every visit, send a single-question push notification asking what they looked for but did not find. Act on the top three answers within 60 days. This group notices when the mall listens.
- **Celebrate personal milestones** — birthdays, anniversaries — with a surprise experiential gift (a restaurant reservation, a spa afternoon) rather than a discount voucher. Money cannot buy the emotional memory that an experience creates.

5.2 Urgent Opportunity — High Income, Low Spending

This group of 3,713 shoppers is the biggest missed opportunity in the dataset — they clearly have the income to spend more, but something is stopping them.

- **Before spending a rupee on promotions, find out why they are holding back.** Place short digital kiosks near exits with a three-question survey: Was there something you wanted but could not find? Did anything make today's visit inconvenient? What would bring you back sooner? The answers will be worth more than any marketing campaign.

Once friction points are identified, create category-specific discovery sessions — a personal styling hour with a fashion store, a cooking demo at the kitchen goods store, a fragrance consultation at the cosmetics counter. High-income shoppers often have untapped interests; the mall just needs to introduce them to the right store.

- **Test an exclusive 'Insider Sale'** — available only to this segment, not the general public — with a 48-hour window and a personalised selection of items curated from stores they have previously browsed but not purchased from. Scarcity and personalisation together can unlock a hesitant buyer.
- **Offer a concierge click-and-collect service:** they select items through an app, a mall assistant collects and bags everything, and it is ready for pickup in 30 minutes. For time-poor, high-income individuals, removing the friction of browsing physical stores may be the entire answer.

5.3 Loyalty Investment — Low Income, High Spending

These 3,741 customers already spend enthusiastically despite not being the wealthiest group, which tells me they have a strong emotional connection to the mall — that is rare and worth protecting carefully.

- **Introduce a zero-cost instalment option** at checkout for purchases above Rs. 2,000. Splitting a Rs. 4,000 bill into three monthly payments removes the affordability ceiling without requiring any subsidy from the retailer. This directly converts intent into purchase.
- **Create a peer referral scheme where existing customers earn loyalty points** — redeemable as mall credits, not just discounts — for every new visitor they bring in. This segment is socially connected and enthusiastic, which makes them natural advocates.
- Time flash sales and limited-period bundle offers to coincide with salary credit dates, typically the 1st and 15th of the month, and around major Indian festivals like Navratri, Diwali, and Eid. The spending willingness is already there — timing the offer correctly amplifies it.
- Publicly acknowledge top spenders in this segment with a 'Loyal Shopper of the Month' display near the food court entrance. A small gesture of recognition costs almost nothing and builds the kind of belonging that makes a customer choose this mall over a competitor every single time.

5.4 Long-term Growth — Low Income, Low Spending

Winning over these 3,798 shoppers is a slow process, and any strategy that tries to rush it will fail — the priority is building a habit of visiting, and purchases will follow over time.

- **Give people reasons to be in the mall even when they are not buying.** Free Wi-Fi zones, weekend children's workshops, community fitness classes in the atrium, and evening cultural performances all build the routine of visiting. A shopper who visits every weekend, even without buying, is far more likely to make a spontaneous purchase than one who visits twice a year.
- **Set the first loyalty reward at a very low threshold** — spend Rs. 300 and unlock a Rs. 50 food court credit. The first redemption is the psychological breakthrough; once a customer has experienced a reward, the habit loop begins.
- **Stock visually appealing, quality-looking products** at price points under Rs. 500 near the main entrances and food court. This segment responds to aspiration — if they can touch and feel something that feels premium but is actually affordable, the first purchase happens naturally.
- **Track this segment's spending score over two consecutive quarters.** Any customer who crosses the median threshold of 51 should automatically be moved into the Low Income High Spending track so their growing momentum receives the right recognition and incentives immediately.

Conclusion

Going into this project, I thought segmenting mall customers would confirm what most retail articles say — that income drives spending. It does not, at least not here. The 0.003 correlation figure will stay with me as a reminder that data has a way of disproving the most obvious assumptions. The four segments that emerged from the median split are almost identical in size, which is genuinely rare in customer datasets and means this mall cannot afford to have a favourite — it needs four working strategies running simultaneously.

Action Rank	Target Segment	Core Intervention	Projected Gain
1 — Do Now	HI – HS	Premium concierge + tiered membership	Revenue lift 20–25%
2 — Do Soon	HI – LS	Discovery events + exclusive invitations	Wallet share +15%
3 — Do Steadily	LI – HS	Instalment plans + milestone recognition	Retention rate +18%
4 — Do Patiently	LI – LS	Footfall habits + entry-level rewards	New buyers +10%

My main recommendation to the marketing team is to stop thinking about customers by what they earn and start thinking about them by how they behave. Cluster 4 from the K-Means model — low income, highest spending score in the dataset — is proof that the most valuable shoppers are not always the wealthiest ones. I would also suggest re-running this segmentation every three months using fresh transaction data, because customers move between segments as their life circumstances change. A student today becomes a salaried professional in two years; the mall should know when that shift happens and respond accordingly.

Original analysis authored by **Neha Bhatt**, Business Analyst, MAY 2026 · Source dataset: Shopping Mall Customer Segmentation (15,079 records) · Toolchain: Python 3, pandas, matplotlib, seaborn, reportlab